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# Candidate Elimination Algorithm
# Without CSV File (Google Colab)
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import pandas as pd

# Step 1: Create Dataset
data = {
    'Sky': ['Sunny', 'Sunny', 'Rainy', 'Sunny'],
    'AirTemp': ['Warm', 'Warm', 'Cold', 'Warm'],
    'Humidity': ['Normal', 'High', 'High', 'High'],
    'Wind': ['Strong', 'Strong', 'Strong', 'Strong'],
    'Water': ['Warm', 'Warm', 'Warm', 'Cool'],
    'Forecast': ['Same', 'Same', 'Change', 'Change'],
    'EnjoySport': ['Yes', 'Yes', 'No', 'Yes']
}

df = pd.DataFrame(data)

print("Training Dataset:\n")
print(df)

# Step 2: Separate Features and Target
concepts = df.iloc[:, :-1].values
target = df.iloc[:, -1].values

# Step 3: Candidate Elimination Algorithm
def candidate_elimination(concepts, target):

    specific_h = concepts[0].copy()

    general_h = [['?' for i in range(len(specific_h))]]
                for j in range(len(specific_h))]

    print("\nInitial Specific Hypothesis:")

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print(specific_h)

for i, h in enumerate(concepts):

    if target[i] == "Yes":

        for x in range(len(specific_h)):
            if h[x] != specific_h[x]:
                specific_h[x] = '?'
                general_h[x][x] = '?'

    elif target[i] == "No":

        for x in range(len(specific_h)):
            if h[x] != specific_h[x]:
                general_h[x][x] = specific_h[x]
            else:
                general_h[x][x] = '?'

# Remove duplicate and fully general hypotheses
final_general = []
for g in general_h:
    if g != ['?'] * len(specific_h) and g not in final_general:
        final_general.append(g)

return specific_h, final_general

# Step 4: Run Algorithm
specific, general = candidate_elimination(concepts, target)

print("\nFinal Specific Hypothesis:")
print(specific)

print("\nFinal General Hypothesis:")
for g in general:
    print(g)

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Training Dataset:

	Sky	AirTemp	Humidity	Wind	Water	Forecast	EnjoySport
0	Sunny	Warm	Normal	Strong	Warm	Same	Yes
1	Sunny	Warm	High	Strong	Warm	Same	Yes
2	Rainy	Cold	High	Strong	Warm	Change	No
3	Sunny	Warm	High	Strong	Cool	Change	Yes

Initial Specific Hypothesis:

['Sunny' 'Warm' 'Normal' 'Strong' 'Warm' 'Same']

Final Specific Hypothesis:

['Sunny' 'Warm' '?' 'Strong' '?' '?']

Final General Hypothesis:

['Sunny', '?', '?', '?', '?', '?']

['?', 'Warm', '?', '?', '?', '?']

